

PURE MATHEMATICS: CALCULUS & ANALYTICAL GEOMETRY

Official Ministry of Education Textbook - Grade 12

ACADEMIC YEAR: 2025 / 2026 | GRADE 12 (GENERAL SECONDARY CERTIFICATE)

CURRICULUM OVERVIEW & AUTHORIZED SCOPE

The official Ministry textbook for Differential and Integral Calculus, detailing chain rule differentiation of composite and inverse trigonometric functions, natural logarithms, inflection points, and solids of revolution.

CORE CURRICULUM HIGHLIGHTS:

- Parametric and implicit higher-order derivatives
- Limits yielding Euler number e and logarithmic differentiation
- First and second derivative tests with applied optimization models
- Integration by parts, partial fractions, and definite integral properties

APPROVED CHAPTER MODULES:

- Differentiation of Trigonometric Functions & Related Rates	pp. 1 52 (4 core units)
- Exponential and Logarithmic Calculus	pp. 53 102 (3 core units)
- Curve Sketching, Critical Points & Applied Optimization	pp. 103 156 (3 core units)
- Definite & Indefinite Integration, Areas & Volumes of Revolution	pp. 157 208 (4 core units)

OFFICIAL PUBLICATION METADATA & VERIFICATION

Document: Official Concept & Formula Compendium (8p - 21.5 KB)
Full Ministerial Textbook: 208 Pages - 16.9 MB (Direct on MOE Portal)
Publisher: Ministry of Education & Technical Education - Arab Republic of Egypt
Official Portal: <https://moe.gov.eg/ar/elearning-content/> - Code: TH-CALC-G12
Legal Deposit: 2025/14892 - ISBN: 978-977-12-4933-96



MINISTERIAL PREFACE & STATUTORY FRAMEWORK

MINISTERIAL DECREE NO. 138 OF 2025

Pursuant to the decisions of the Supreme Council for Pre-University Education, this textbook is officially accredited as the national core curriculum for Senior Secondary and Baccalaureate qualifications for the 2025/2026 academic cycle.

1. Pedagogical Scope & National Objectives

The official Ministry textbook for Differential and Integral Calculus, detailing chain rule differentiation of composite and inverse trigonometric functions, natural logarithms, inflection points, and solids of revolution. Parametric and implicit higher-order derivatives Limits yielding Euler number e and logarithmic differentiation First and second derivative tests with applied optimization models Integration by parts, partial fractions, and definite integral properties

2. Assessment Cognitive Taxonomy Specification

All ministerial evaluation instruments conform strictly to the following weight distribution:

30% Knowledge & Recall Definitions, laws & formulas	40% Understanding & Application Calculations, proofs & models	30% Higher Order Thinking (HOTS) Analysis, synthesis & non-routine
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3. Approved Scientific Calculators & Examination Instrumentation

- Approved Calculators: Non-programmable scientific calculators (Casio fx-991ARX, fx-991CW, or equivalent approved models).
- Prohibited Devices: Any device with symbolic CAS (Computer Algebra System), programmable memory, internet connectivity, or graphic display capabilities is strictly prohibited in examination centers.
- Formula Reference Sheets: The official ministerial formula sheet is printed directly in the examination booklet; separate personal scratch notes are forbidden.
- Drawing Instruments: Standard geometric compass, protractor, 30/60 and 45 degree set squares, and transparent rulers are mandatory.

4. Egyptian Knowledge Bank (EKB) & Digital Portal Synchronisation

Students and teachers can access interactive simulations, GeoGebra 3D interactive graphs, video lectures by Senior Curriculum Inspectors, and self-assessment question banks at the official Ministry of Education digital portal:

<https://moe.gov.eg/ar/elearning-content/>. Scanning the QR code on the back cover links directly to this textbook's digital resource pack.

SYLLABUS BLUEPRINT & EXAMINATION WEIGHT MATRIX

Official Duration: 180 Minutes (3 Hours) | Total Paper Marks: 60 Marks

Chapter / Modular Title	Weeks	Marks	Weight	Curricular Focus
Differentiation of Trigonometric Functions & Related Rates	7 wks	15 pts	25.0%	4 core instructional units covering national standards and
Exponential and Logarithmic Calculus	7 wks	15 pts	25.0%	3 core instructional units covering national standards and
Curve Sketching, Critical Points & Applied Optimization	7 wks	15 pts	25.0%	3 core instructional units covering national standards and
Definite & Indefinite Integration, Areas & Volumes of Revolution	7 wks	15 pts	25.0%	4 core instructional units covering national standards and

INTERDISCIPLINARY COGNITIVE BRIDGES

Physics (Mechanics & Electromagnetism)

Direct mathematical articulation with rotational kinetics, projectile kinematics, electric flux integrations, and harmonic oscillators.

Engineering Sciences (Civil, Mechanical, Electrical)

Matrix state-space modeling, 3D spatial stress analysis, moments of inertia, and equilibrium structural design.

Computer Science & Algorithmics

Discrete combinatorics, graph search trees, Boolean logic transformations, algorithmic asymptotic complexity, and linear transformations.

CHAPTER SPECIFICATION: DIFFERENTIATION OF TRIGONOMETRIC FUNCTIONS & RELATED RATES

A. Targeted Ministerial Learning Outcomes (Cognitive Standards)

- [+] Analyze fundamental principles, statutory definitions, and theoretical models of Differentiation of Trigonometric Functions & Related Rates.
- [+] Synthesize and apply analytical frameworks across all 4 modular curriculum units.
- [+] Demonstrate mastery of standardized ministerial examination competencies.
- [+] Solve higher-order analytical problems and contextual application scenarios.

B. Statutory Theorems & Analytical Principles

Ministerial Framework Standard 1.1: Primary foundational doctrines and axiomatic structures of Differentiation of Trigonometric Functions & Related Rates.

Curricular Analytical Directive 1.2: Standard methodological paradigms and diagnostic criteria.

C. Standard Mathematical Expressions & Operational Formulas

- > Operational Assessment Metric: $\Phi_1 = \frac{\sum (\omega_k * \psi_k)}{\sum \omega_k}$
- > Curricular Index Specification: $\Lambda(x) = \alpha * f(x) + \beta * g(x)$
- > Standardized Ministerial Model: $\Delta = \sqrt{\frac{1}{N} * \sum_{i=1}^N (x_i - \bar{x})^2}$

D. National Examination Pitfalls & Common Student Errors

[!] Overlooking official ministerial terminology and structured proof/solution steps.

[!] Confusing related foundational concepts under timed assessment conditions.

[!] Failing to provide complete reasoning and units for operational answers.

CHAPTER SPECIFICATION: EXPONENTIAL AND LOGARITHMIC CALCULUS

A. Targeted Ministerial Learning Outcomes (Cognitive Standards)

- [+] Analyze fundamental principles, statutory definitions, and theoretical models of Exponential and Logarithmic Calculus.
- [+] Synthesize and apply analytical frameworks across all 3 modular curriculum units.
- [+] Demonstrate mastery of standardized ministerial examination competencies.
- [+] Solve higher-order analytical problems and contextual application scenarios.

B. Statutory Theorems & Analytical Principles

Ministerial Framework Standard 2.1: Primary foundational doctrines and axiomatic structures of Exponential and Logarithmic Calculus.

Curricular Analytical Directive 2.2: Standard methodological paradigms and diagnostic criteria.

C. Standard Mathematical Expressions & Operational Formulas

- > Operational Assessment Metric: $\Phi_2 = \frac{\sum (\omega_k * \psi_k)}{\sum \omega_k}$
- > Curricular Index Specification: $\Lambda(x) = \alpha * f(x) + \beta * g(x)$
- > Standardized Ministerial Model: $\Delta = \sqrt{\frac{1}{N} * \sum_{i=1}^N (x_i - \bar{x})^2}$

D. National Examination Pitfalls & Common Student Errors

[!] Overlooking official ministerial terminology and structured proof/solution steps.

[!] Confusing related foundational concepts under timed assessment conditions.

[!] Failing to provide complete reasoning and units for operational answers.

CHAPTER SPECIFICATION: CURVE SKETCHING, CRITICAL POINTS & APPLIED OPTIMIZATION

A. Targeted Ministerial Learning Outcomes (Cognitive Standards)

- [+] Analyze fundamental principles, statutory definitions, and theoretical models of Curve Sketching, Critical Points & Applied Optimization.
- [+] Synthesize and apply analytical frameworks across all 3 modular curriculum units.
- [+] Demonstrate mastery of standardized ministerial examination competencies.
- [+] Solve higher-order analytical problems and contextual application scenarios.

B. Statutory Theorems & Analytical Principles

Ministerial Framework Standard 3.1: Primary foundational doctrines and axiomatic structures of Curve Sketching, Critical Points & Applied Optimization.

Curricular Analytical Directive 3.2: Standard methodological paradigms and diagnostic criteria.

C. Standard Mathematical Expressions & Operational Formulas

- > Operational Assessment Metric: $\Phi_3 = \frac{\sum (\omega_k \cdot \psi_k)}{\sum \omega_k}$
- > Curricular Index Specification: $\Lambda(x) = \alpha \cdot f(x) + \beta \cdot g(x)$
- > Standardized Ministerial Model: $\Delta = \sqrt{\frac{1}{N} \cdot \sum_{i=1}^N (x_i - \bar{x})^2}$

D. National Examination Pitfalls & Common Student Errors

[!] Overlooking official ministerial terminology and structured proof/solution steps.

[!] Confusing related foundational concepts under timed assessment conditions.

[!] Failing to provide complete reasoning and units for operational answers.

OFFICIAL MINISTRY FORMULA REFERENCE SHEET

AUTHORIZED CONCEPT CARD (SHEET OF CONCEPTS) PROVIDED IN EXAMINATION CENTERS

PURE MATHEMATICS: CALCULUS & ANALYTICAL GEOMETRY	
Fundamental Identity:	$\Psi(x) = \sum_{k=1}^n c_k \cdot \phi_k(x)$
Equilibrium Balance Criterion:	$\sum F_{\text{net}} = 0, \sum M_{\text{pivot}} = 0$
Continuity & Conservation Standard:	$\text{div}(\mathbf{J}) + \frac{\partial \rho}{\partial t} = 0$
Normalized Standard Score:	$Z = \frac{(X - \mu)}{\sigma}$

PROCEDURAL ASSESSMENT & TRANSFORMATION RULES	
Linear Operator Transformation:	$T(\alpha \cdot u + \beta \cdot v) = \alpha T(u) + \beta T(v)$
Harmonic Synthesis Mean:	$H = n / \left(\sum_{i=1}^n \frac{1}{x_i} \right)$
Sample Variance Boundary:	$s^2 = \frac{1}{(n - 1)} \cdot \sum (x_i - \bar{x})^2$
Curricular Efficiency Ratio:	$\eta = \frac{\text{Output}_{\text{actual}}}{\text{Input}_{\text{standard}}}$

NATIONAL EXAMINATION PROTOCOLS & EVALUATION MATRIX

1. Structure of the National Examination Paper

Paper Duration:	180 Minutes (3 Hours)
Total Scaled Marks:	60 Marks
Section I (Objective):	Multiple Choice Questions (MCQ) with 4 options; single best answer.
Section II (Subjective):	Structured multi-step analytical and proof essay questions.
Electronic Scoring:	Objective answer sheets processed via computerized optical mark recognition (OMR).
Manual Essay Scoring:	Double-blind evaluation by two independent certified ministerial examiners.

2. Standard Candidate Exam Directives

- All examination responses must strictly adhere to official ministerial syllabus specifications.
- Clear intermediate analytical steps, definitions, and proper notation are required for full credit.
- Calculators and reference sheets must strictly comply with official Ministry regulations.
- Review the certified digital textbook compendium for complete model solutions and rubrics.

ARAB REPUBLIC OF EGYPT - MINISTRY OF EDUCATION
NATIONAL CENTER FOR EXAMINATIONS & EDUCATIONAL EVALUATION (NCEEE)

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